

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE:

STATION:	SALEM		
SYSTEM:	Auxiliary Feedwater		
TASK:	Perform Auxiliary Feedwater Back Leakage Test		
TASK NUMBER:	N0610160204		
JPM NUMBER:	15-01 NRC IP-j		
ALTERNATE PATH:	☐	K/A NUMBER:	061 K1.02
APPLICABILITY:		IMPORTANCE FACTOR:	
EO ☐	RO ☒	STA ☐	SRO ☒
		RO	3.4
		SRO	3.7
EVALUATION SETTING/METHOD:	Inplant- Simulate		
REFERENCES:	S1.OP-PT.AF-0002, Rev. 7, Auxiliary Feedwater Backleakage (checked 8-5-16)		
TOOLS AND EQUIPMENT:	Calculator		
VALIDATED JPM COMPLETION TIME:	5 minutes		
TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS:	N/A		
Developed By:	G Gauding Instructor	Date:	8-5-16
Validated By:	S Bickhart / J Hancock SME or Instructor	Date:	9-30-16
Approved By:	 Training Department	Date:	10/10/16
Approved By:	 Operations Department	Date:	10/10/16
ACTUAL JPM COMPLETION TIME:			
ACTUAL TIME CRITICAL COMPLETION TIME:			
PERFORMED BY:			
GRADE: ☐ SAT ☐ UNSAT			
REASON, IF UNSATISFACTORY:			
EVALUATOR'S SIGNATURE:		DATE:	

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE:

NAME: _____

DATE: _____

SYSTEM: Auxiliary Feedwater System

TASK: Perform Auxiliary Feedwater Back Leakage Test

TASK NUMBER: N0610160204

INITIAL CONDITIONS:

Unit 1 is operating at 100% power. 11 AFW pump was secured 30 minutes ago after its normally scheduled quarterly surveillance run.

INITIATING CUE:

You are directed to perform S1.OP-PT.AF-0002, Auxiliary Feedwater Backleakage.

Successful Completion Criteria:

1. All critical steps completed.
2. All sequential steps completed in order.
3. All time-critical steps completed within allotted time.
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made.

Task Standard for Successful Completion:

1. Determine 11 AFW pump and piping requires venting.
2. Vents 11 AFW pump and piping.

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____
DATE: _____

SYSTEM: AFW

TASK: Perform Auxiliary Feedwater Back Leakage Test

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
		Provide operator with blank copy of S1.OP-PT.AF-0002, Rev. 7, and a calculator.	Reviews and initial all Prerequisites and Precautions and Limitations.		
* * *	5.1.1	RECORD following on Attachment 1, Section 1.0: • AFW Line Temperatures (1TA16555I) • Highest line temperature • Average line temperature • Temperature difference	After locating 1TA165551, provide the following AFW line temperatures: 11- 104 12- 106 13- 161 14- 108 On Attachment 1, records AFW temperatures, calculates total of 479 , divides that by 4 and determines highest line temp is 161 and average line temp is 119.75 . Determines Temp Diff 161- 119.75 = 41.25 .		
	5.1.2	IF temperature in each line is <120°F, AND temperature difference between highest and average line temp is ≤10°F, THEN RECORD Test Results in Attachment 1, Section 5.0.	Determines highest temp is >120°F		

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JOB PERFORMANCE MEASURE**

NAME: _____
DATE: _____

SYSTEM: AFW

TASK: Perform Auxiliary Feedwater Back Leakage Test

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
	5.1.3	IF temperature in any line is $\geq 120^{\circ}\text{F}$ AND $< 160^{\circ}\text{F}$, THEN: A. RECORD Auxiliary Feedwater System piping temperatures every hour in Control Room Narrative Log. B. RECORD Test Results in Attachment 1, Section 5.0.	Determines highest temp is $> 160^{\circ}\text{F}$		
*	5.1.4	IF temperature difference between highest and average line temp is $> 10^{\circ}\text{F}$, THEN RECORD Test Results in Attachment 1, Section 5.0.	Determines temperature difference between highest and average line temp is $> 10^{\circ}\text{F}$, and records Test Results in Attachment 1, Section 5.0., as UNSAT for all three categories.		
	5.1.6	IF temperature in any line is $\geq 160^{\circ}\text{F}$ THEN: A. RECORD findings in Control Room Narrative Log. B. VENT Auxiliary Feedwater pumps/piping IAW Section 5.2.			

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____
DATE: _____

SYSTEM: AFW

TASK: Perform Auxiliary Feedwater Back Leakage Test

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
*	5.2.1	IF 11 Auxiliary Feedwater Pump and piping is affected, THEN: A. SLOWLY THROTTLE OPEN 11AF83, AF PUMP VENT, until a solid stream of water is observed. B. CLOSE 11AF83. C. SLOWLY THROTTLE OPEN 11AF114, AF PUMP DISCH VENT, until a solid stream of water is observed. D. CLOSE 11AF114.	Locates and opens 11AF83, AF PUMP VENT, until a solid stream of water is observed. Cue: A solid stream of water is observed. Closes 11AF83 Locates and opens 11AF114, AF PUMP DISCH VENT, until a solid stream of water is observed. Cue: A solid stream of water is observed. Closes 11AF114.		
			Terminate JPM		

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JOB PERFORMANCE MEASURE

JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

- ✓ 1. Task description and number, JPM description and number are identified.
- ✓ 2. Knowledge and Abilities (K/A) references are included.
- ✓ 3. Performance location specified. (in-plant, control room, or simulator)
- ✓ 4. Initial setup conditions are identified.
- ✓ 5. Initiating and terminating Cues are properly identified.
- ✓ 6. Task standards identified and verified by SME review.
- ✓ 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).
- ✓ 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. 7 Date 1/6/10
- 2 ✓ 9. Pilot test the JPM:
a. verify Cues both verbal and visual are free of conflict, and
b. ensure performance time is accurate.
- _____ 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.
- _____ 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor: [Signature] HANCOCK

Date: 9/30/16

SME/Instructor: [Signature] BICKHART

Date: 9/30/16

SME/Instructor: _____

Date: _____

JOB PERFORMANCE MEASURE

INITIAL CONDITIONS:

Unit 1 is operating at 100% power. 11 AFW pump was secured 30 minutes ago after its normally scheduled quarterly surveillance run.

INITIATING CUE:

You are directed to perform S1.OP-PT.AF-0002, Auxiliary Feedwater Backleakage.